

Numerical Analysis PhD Exam, August 1995

1. Consider the Householder matrix $H = I - 2ww^T$, where $w^T w = 1$.
 - (a) Show that H is orthogonal.
 - (b) Determine the eigenvalues of H .
 - (c) Given a vector x and an integer k , find a vector w such that $(Hx)_i = x_i$ for $i < k$ and $(Hx)_i = 0$ for $i > k$.
 - (d) Explain why one needs to be careful about the choice of sign in the formula for w .
2. Let A be a symmetric, positive definite matrix. Show that if A is LU factored, then the elements of U in absolute value are bounded by the largest diagonal element of A .
3. Give a careful statement and proof of Gershgorin's Theorem. Find upper and lower bounds for the set of eigenvalues for the matrix

$$\begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}.$$

4. Let A be a nonsingular $n \times n$ matrix and $\{X_k\}$ a sequence of $n \times n$ matrices generated by the iteration

$$X_{k+1} = X_k + X_k(I - AX_k).$$

Show that if X_0 is chosen such that the spectral radius of $I - AX_0$ is strictly less than 1, then $\{X_k\}$ converges to the inverse of A .

5. For a given partition $\{x_0, x_1, \dots, x_n\}$ of an interval $[a, b]$ and function f defined on $[a, b]$:
 - (a) State the divided difference formula for the polynomial $p_n(x)$ interpolating f from the data $(x_k, f(x_k))$, $k = 0, 1, \dots, n$.
 - (b) Show that the error in the approximation in part (a) is given by

$$f(x) - p_n(x) = (x - x_0)(x - x_1) \cdots (x - x_n) f[x_0, x_1, \dots, x_n].$$

- (c) For $\{x_0, x_1, x_2\} = \{0, 1, 2\}$ and $f(x) = 1/(1 + x^2)$, compute the divided difference formula for $p_2(x)$.
6. Let A be a real, positive definite $n \times n$ matrix and b be an $n \times 1$ vector.
 - (a) Derive the steepest descent algorithm for approximating the true solution x of $Ax = b$, by a sequence of iterates $\{x_k\}$.
 - (b) Define the real-valued function h on $\mathbb{R}^n$ by

$$h(x) = (Ax - b)^T A^{-1} (Ax - b)$$

and show that

$$h(x_{k+1}) \leq (1 - c(A)^{-1})^2 h(x_k),$$

where $c(A)$ is the condition number of the matrix A .

7. For an algorithm, define the terms inherent error, total effect of rounding, and numerically stable. Carefully formulate a numerically stable algorithm for computing the positive solution of the quadratic equation $x^2 + 2px - q = 0$, where p and q are arbitrary positive numbers. Justify the numerical stability of your algorithm.
8. Give a careful proof that the DFT of a convolution of two finite sequences (not necessarily of the same length) is proportional to the product of the DFT's of the original sequences, suitably padded.
9. Given the N sampled values $f(2k\pi/N)$, $k = 0, 1, 2, \dots, N - 1$, of a twice continuously differentiable, 2π -periodic function f , outline a method of computing the Fourier coefficients of f which has the following properties:
- makes use of the FFT routine;
 - produces an estimate which improves as $N \rightarrow \infty$.

Estimate the accuracy of your method. If possible, state an explicit formula for its implementation.